



Sanchai

NEPALI CLINICAL LEDGER

Zero-Hallucination Nepali Clinical Health Ledger & Offline-First AI

Grounding Multimodal Health Records in Nepal's Real-World Clinical Ecosystem

■ 0% Hallucinations

Verified 300-concept clinical ontology

■ Dual-Tier Gemma 4

Cloud (31B) +
Edge Local Fallback (2B QAT)

■ Human Approval Gate

Clinician review before
ledger commit

■ Emergency Triage QR

Sub-second offline paramedic
card

• PITCH DECK: 3-MINUTE PRESENTATION + 2-MINUTE LIVE DEMONSTRATION

Powered by Google Gemma 4

Four Fatal Flaws in Nepal's Healthcare Flow

Why generic digital health apps and unconstrained chatbots fail in real-world clinical practice

CRITICAL RISK

The 'Plastic Bag' Records

Faded thermal receipts, handwritten cursive scripts, and fragmented paper slips scattered across clinics. When patients visit a new hospital, past medical history is completely lost.

FATAL RISK

The Brand vs. Generic Trap

Pharmacies dispense commercial brands. A patient taking Cetamol in Kathmandu is prescribed Paracetamol in Pokhara, causing accidental double dosing and acute liver toxicity.

HIGH COMPLEXITY

Trilingual & Code-Mixed Notes

Real notes mix Devanagari, Romanized phonetics ('chhati polyo', 'jwaro chhaina'), and medical shorthand. Generic English LLMs miss cultural idioms and misdiagnose symptoms.

TERRAIN BARRIER

The Mountain Connectivity Barrier

Landslides, floods, and power outages regularly cut fiber lines to rural health posts (Swasthya Chauki). Cloud-only health apps become completely useless when offline.

The Sanchai Zero-Hallucination Pipeline

Deterministic clinical scaffolding binding Gemma 4 to verified ontology ground truth

Deterministic 3-Tier Clinical Normalization Engine

Tier 1: Sub-Millisecond Exact Match

0 ms

O(1) hash table lookup across 300 concepts. Matches Devanagari, English, Romanized aliases, and DDA commercial brands directly. Zero hallucinations possible.

Tier 2: Orthographic & Matra Fuzzy Match

< 2 ms

RapidFuzz bounded Levenshtein distance. Recovers OCR-corrupted vowel signs ('jwarou' -> 'jwaro') without semantic drift or concept confusion.

Tier 3: Gemma 4 Constrained Guardrail

AI Guard

When ambiguous phrasing is encountered, Gemma 4 is presented with allowed candidates. It must choose an existing ID or output UNKNOWN. Negative rejection enforced.

Four Safety Guardrails

300 Verified Concepts

Sourced directly from Nepal DDA, NLEM 2021, and WHO ICD-11.

Directional Negation

'jwaro chhaina' (no fever) is captured with 100% accuracy and excluded.

Brand -> Generic Resolution

Taxim-O -> Cefixime; prevents dangerous duplicate therapies.

Approval-Before-Write Gate

Human clinician reviews and commits before saving to record.

NepClinBench Live Evaluation & Ablation

Rigorous benchmark on 60 real bilingual clinical test cases comparing Sanchai vs. Raw Gemma 4

EVALUATION METRIC	SANCHAI 3-TIER PIPELINE	RAW GEMMA 4 BASELINE	SANCHAI ADVANTAGE
Exact Set Match (N=60 Gold Cases)	96.7% (58 / 60)	46.7% (28 / 60)	+50.0% GAIN
Precision (Zero Hallucination Guarantee)	1.000 (0% Hallucination)	0.817 (18.3% Hallucinated)	+22.4 PTS
Brand to Generic Resolution (DDA)	100.0% (54 / 54)	41.7% (23 / 54)	+58.3% GAIN
Directional Negation Accuracy (Pertinent Negatives)	100.0% (10 / 10)	50.0% (5 / 10)	+50.0% GAIN
Duration & Dosage Parsing	100.0% (12 / 12)	66.7% (8 / 12)	+33.3% GAIN
FHIR R4 Bundle Validity	100% (Strict Valid JSON)	0% (Unstructured English)	100% INTEROPERABLE
Deterministic Edge Latency	Yes (< 5 ms offline)	No (> 800 ms cloud API)	160x FASTER

End-to-End Continuity: SanchAI & Emergency QR

Empowering patients at home, assisting doctors in OPD, and saving lives in ambulances

SanchAI Copilot

EHR GROUNDED ASSISTANT

- **Patient Context Aware**
Knows active conditions, allergies, and ongoing medications completely.
- **Multimodal Ingestion**
Uploads prescription photos and lab PDFs for instant interpretation.
- **Plain Nepali Explanations**
Translates CBC, Widal, and lipid panel jargon into everyday Nepali.
- **Pre-Visit Doctor Briefing**
One-click synthesis preparing complaints and questions for doctors.

Critical Safety Guardrails

ALLERGY RISK DEFENSE

- **Allergy Cross-Referencing**
Checks user query against documented patient allergies automatically.
- **Beta-Lactam Protection**
Flags Amoxicillin/Augmentin if patient has documented Penicillin allergy.
- **Zero Autonomous Writes**
Cannot mutate authoritative health ledger without clinician approval.
- **Doctor Summary A4 PDF**
One-click print-ready clinical report with embedded QR code verification.

Emergency Optical QR

AMBULANCE TRIAGE IN < 1S

- **High-Contrast Segno QR**
Encodes deterministic token readable by any phone camera or optical scanner.
- **Offline Paramedic View**
Instant triage strip: Blood group, severe allergies, chronic diagnoses.
- **Unconscious Patient Care**
Paramedics make life-saving decisions before reaching hospital doors.
- **HL7 FHIR R4 Export**
Standardized interoperability for hospital EHRs (Bahmni, Epic, Cerner).

Why Offline Architecture is Life or Death in Nepal

A digital health system in Nepal that fails when the internet cuts out is a health system that fails when needed most

"A digital health system in Nepal that fails when the internet cuts out
is a health system that fails when it is needed most."

LAYER 1

< 5 ms Latency

Deterministic Rule Core

100% Offline & Local

Exact dictionary matching and
fuzzy matra parsing run locally
with zero network requirement.
Instant clinical lookup even on
battery power.

LAYER 2

High Nuance

Cloud Acceleration

Online Gemma 4 (31B)

When connected to internet,
routes to gemma4:31b-cloud for
complex prescription handwriting
OCR, deep reasoning, and doctor
summaries.

LAYER 3

Zero Downtime

Automatic Edge Fallback

Offline Gemma 4 (2B QAT)

If fiber line cuts or mobile data
drops, Sanchai seamlessly
switches to locally installed
gemma4:e2b-it-qat running on the
local device.

Sustainable Healthcare Infrastructure for Nepal

Transforming fragmented plastic bag records into permanent health agency for every citizen

Sustainable B2B2C Healthcare Model

■ Hospital & Clinic OPD SaaS

Hospitals pay a modest monthly license for the ingestion studio, automated FHIR conversion, and doctor summary generation,

■ National Health Insurance Board

Partnering with Swasthya Bima Board (Health Insurance Board) to detect duplicate drug dispensing and stop fraudulent claims across

■ Free Personal Ledger for Citizens

Personal records, emergency QR cards, and SanchAI assistance remain completely free for citizens to ensure equitable access.

Roadmap & Future Scope

● Lexicon Scale

Expanding from 300 to 1,500+ DDA concepts covering tertiary oncology and pediatrics.

● Ministry of Health Integration

Direct API integration with national DHIS2 and digital health ID.

● Pharmacy Barcode Scanners

Equipping community pharmacies to verify prescriptions before dispensing.

Sanchai Hunuhunchha?

Turning plastic bags into longitudinal health agency.

Thank you! Live Demo starting now.